

Sample Calculations

Reference: P. Wankat, "Flash Distillation," in *Separation process engineering: Includes Mass transfer analysis*, 4th ed., Pearson Education, 2017, pp. 25–30.

The volatility α_{AB} is given by the following equation, where A represents the more volatile component (ethanol) and B represents the less volatile component (water). We can calculate volatility α_{AB} in both the distillate and the bottoms and use the values to calculate the average relative volatility:

$$\alpha_{AB(Dist)} = \left(\frac{K_A}{K_B}\right)_{Dist} \quad \alpha_{AB(Bot)} = \left(\frac{K_A}{K_B}\right)_{Bot}$$

Where:

K_A for ethanol in distillate and bottoms is:

$$(K_A)_{Dist} = \left(\frac{y_A}{x_A}\right)_{Dist} \quad (K_A)_{Bot} = \left(\frac{y_A}{x_A}\right)_{Bot}$$

K_B for water in distillate and bottoms:

$$(K_B)_{Dist} = \left(\frac{y_B}{x_B}\right)_{Dist} \quad (K_B)_{Bot} = \left(\frac{y_B}{x_B}\right)_{Bot}$$

The average relative volatility is $(\alpha_{AB})_{avg} = \sqrt{\alpha_{AB(Dist)}\alpha_{AB(Bot)}}$

Example:

Distillate (condensate)

Mole fraction of ethanol in distillate (condensate) calculated from v/v % measurements:

$$x_{A(dist)} = 0.7569$$

Use VLE data to find $y_{A(dist)} \rightarrow 0.794$

Calculate mole fractions x_B and y_B of water in distillate:

x_{H_2O} in distillate:

$$\begin{aligned} x_{H_2O} &= 1 - x_{EtOH} \\ x_{H_2O} &= 1 - .7569 = .2431 \end{aligned}$$

y_{H_2O} in distillate:

$$\begin{aligned} y_{H_2O} &= 1 - y_{EtOH} \\ y_{H_2O} &= 1 - .794 = 0.206 \end{aligned}$$

K_A for ethanol in distillate:

$$(K_A)_{Dist} = \left(\frac{y_A}{x_A}\right)_{Dist} = \frac{.794}{.7569} = 1.049$$

$(K_B)_D$ for water in distillate:

$$(K_B)_{Dist} = \left(\frac{y_B}{x_B}\right)_{Dist} = \frac{.2060}{.2431} = 0.8474$$

Volatility for distillate $(\alpha_{AB})_{Dist}$:

$$(\alpha_{AB})_D = \left(\frac{K_A}{K_B}\right)_D = \frac{(K_A)_D}{(K_B)_D} = \frac{1.049}{.8474} = 1.238$$

Mole fraction of ethanol for the bottom sample calculated from v/v% measurements:

$$x_{A(Bot)} = 0.035$$

From the VLE equilibrium data:

$$(y_A)_{Bot} = 0.282 \text{ at } (x_{EtOH})_{Bot} = 0.035$$

K_A for ethanol in bottom sample:

$$(K_A)_{Bot} = \left(\frac{y_A}{x_A} \right)_{Bot} = \frac{.282}{.0350} = 8.066$$

x_{H_2O} in bottom sample:

$$\begin{aligned} x_{H_2O} &= 1 - x_{EtOH} \\ x_{H_2O} &= 1 - .0350 = .9650 \end{aligned}$$

y_{H_2O} in bottom sample:

$$\begin{aligned} y_{H_2O} &= 1 - y_{EtOH} \\ y_{H_2O} &= 1 - 0.282 = .7180 \end{aligned}$$

$(K_B)_{Bot}$ for water in bottom sample:

$$(K_B)_{Bot} = \left(\frac{y_B}{x_B} \right)_{Bot} = \frac{.7180}{.9650} = 0.7440$$

Volatility for bottom sample $(\alpha_{AB})_{Bot}$:

$$(\alpha_{AB})_{Bot} = \left(\frac{K_A}{K_B} \right)_{Bot} = \frac{(K_A)_{Bot}}{(K_B)_{Bot}} = \frac{8.066}{.7440} = 10.841$$

Average relative volatility:

$$(\alpha_{AB})_{avg} = \sqrt{\alpha_{Dist} \alpha_{Bot}} = \sqrt{(1.238)(10.841)} = 3.663$$

Fenske Equation for theoretical number of plates:

$$\begin{aligned} N + 1 &= \frac{\log \left(\left(\frac{x_{EtOH}}{x_{H_2O}} \right)_d \left(\frac{x_{H_2O}}{x_{EtOH}} \right)_b \right)}{\log \left(\alpha_{\frac{EtOH}{H_2O}} \right)_{avg}} \\ N &= \frac{\log \left(\left(\frac{.7569}{.2431} \right) \left(\frac{.0350}{.9650} \right) \right)}{\log(3.663)} - 1 = 2.455 \end{aligned}$$

Column Efficiency:

$$Efficiency (\%) = \frac{\text{theoretical plates}}{\text{actual plates}} * 100$$

Number of actual plates = 9 plates.

$$Efficiency (\%) = \frac{2.455}{9} * 100 = 27.28 \%$$